# HMM-based chromatin state segmentation on chr22 using three histone marks (H3K4me3/H3K27ac/H3K36me3)

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**Continuous ChIP-seq bigWig tracks provide rich epigenomic information but are not readily interpretable as discrete regulatory annotations. We partitioned human chr22 into 200-bp bins, applied log1p transformation and per-mark z-score standardization, and performed unsupervised segmentation with a multivariate Gaussian hidden Markov model trained by EM (Baum–Welch) and decoded by Viterbi. The model identified four recurrent chromatin states with distinct emission signatures across H3K4me3, H3K27ac, and H3K36me3. These states showed biologically consistent enrichments at CpG islands, transcription start sites, and gene bodies, and diagnostic analyses highlighted practical failure modes and safeguards for robust state-number selection.**

chromatin state | hidden Markov model | ChIP-seq | histone marks | segmentation

Chromatin organization is commonly profiled by ChIP-seq as genome-wide coverage tracks that are stored and shared as continuous signals (e.g., bigWig)(1).These tracks are ideal for visualization and for detecting prominent peaks, yet they do not directly provide a compact, discrete annotation of regulatory function along the genome. Peak-centric analyses can identify sharp, high-confidence binding or modification events, but many biologically meaningful patterns are broader than a narrow peak (e.g., promoter-proximal plateaus, transcription-associated gene-body enrichment, or extended regulatory neighborhoods). In addition, local technical variation and biological heterogeneity can produce bin-to-bin fluctuations that blur boundaries when the genome is interpreted pointwise. A segmentation framework addresses these issues by converting dense, continuous tracks into a sequence of contiguous genomic segments with shared latent "states," enabling summary statistics at the state level and principled validation against independent annotations such as CpG islands, transcription start sites (TSS), and gene bodies(2).

We focused on three widely used histone marks that together span major regulatory contexts and provide an interpretable basis for unsupervised state discovery. H3K4me3 is typically enriched at active promoters and tightly localized around TSS, making it a canonical promoter-associated signal(3).H3K27ac marks active regulatory elements and is commonly observed at enhancers as well as active promoter regions, capturing regulatory activity beyond promoter cores(4). H3K36me3 is enriched over transcribed gene bodies and reflects transcription-coupled chromatin states associated with elongation(3).Importantly, these marks are not redundant: promoter-proximal and distal regulatory activity can share acetylation yet differ in H3K4me3, and gene-body enrichment of H3K36me3 can occur with relatively weak promoter marks. Modeling them jointly therefore allows states to be defined by combinations of signals—promoter-like, enhancer-like/weak-regulatory, gene-body-like, and background-like—rather than by any single track.

To capture both the multivariate structure of these signals and their spatial continuity along chromosomes, we used a hidden Markov model (HMM)(5, 6).In an HMM, each genomic bin is generated from an unobserved state, and adjacent bins are coupled through a first-order Markov transition process. This design has two practical advantages for chromatin segmentation. First, it provides an explicit mechanism for smoothing: state self-transition probabilities favor locally stable domains and prevent label "flickering" caused by local noise. Second, it yields a probabilistic framework in which state emission parameters summarize characteristic mark patterns, while decoding produces contiguous segments that can be directly intersected with external genomic annotations for biological interpretation. Together, these properties make HMMs a natural choice for converting noisy, continuous ChIP-seq tracks into interpretable, domain-like chromatin state maps.

In this study, we apply a multivariate Gaussian HMM to three normalized histone-mark tracks on human chr22 (200-bp bins) to infer four recurrent chromatin states, and we assess their interpretability using enrichment at CpG islands, TSS, and gene bodies together with diagnostic checks that guard against unstable state-number selection.

## Significance Statement

Genome-wide epigenomic tracks are typically continuous signals that are hard to translate into discrete functional annotations. We segment chr22 into contiguous chromatin states using a multivariate Gaussian hidden Markov model trained on three widely used histone marks (H3K4me3, H3K27ac, and H3K36me3). The model converts noisy, bin-level variation into domain-like segments and yields four interpretable states that show consistent enrichments at CpG islands, transcription start sites, and gene bodies. This workflow provides a compact, reproducible way to summarize chromatin organization from continuous ChIP-seq signals and highlights diagnostic checks that reveal when model selection can become unstable on small or low-contrast regions.

HaiFeng Q. designed the analysis, implemented the workflow, and wrote the manuscript.

The author declares no competing interest.

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## Results

**Preprocessing harmonizes mark scales and reveals complementary input structure.** Raw binned signals on chr22 show strongly right-skewed, long-tailed distributions for all three marks, indicating that a small fraction of extreme peaks can dominate Gaussian-parameter estimation if modeled on the original scale (Fig. 1A). After applying log1p and per-mark z-score standardization, the densities become substantially more concentrated and comparable across marks (Fig. 1B), and genome-view examples illustrate that normalized tracks reduce scale disparities and make broad, low-to-moderate domains visually resolvable rather than being visually overwhelmed by a few extreme peaks (Fig. 1C–D). Importantly, this transformation is used as a robustness step to reduce leverage from extremes and to place marks on commensurate scales, rather than to enforce perfect normality.

Beyond scale harmonization, the three normalized marks retain complementary structure rather than collapsing into a near-collinear signal. Pairwise correlations are moderate (H3K4me3–H3K27ac = 0.53; H3K4me3–H3K36me3 = 0.45; H3K27ac–H3K36me3 = 0.59; Fig. 2), consistent with partial information overlap but substantial non-redundant variation. This combination—reduced extreme-value dominance and moderate inter-mark correlation—supports multivariate state discovery in which latent states reflect distinct combinations of promoter-, enhancer-, and transcription-associated patterns rather than any single track alone(5, 6).

These preprocessing and correlation results motivate fitting a multivariate HMM on the normalized 3D signal vectors. In the next section, we show that the fitted model recovers a small number of recurrent emission programs with clear separation on the z-score scale, providing a compact summary of chr22 chromatin organization.

**A four-state HMM captures distinct multivariate emission programs.** Fitting a multivariate Gaussian HMM to the normalized three-mark vectors yields four recurrent states with clearly stratified emission means (Fig. 3), providing a compact representation of chr22 chromatin variation in a three-dimensional mark space(7, 8). The inferred states form an ordered continuum from low signal to high composite activity. One state exhibits uniformly low mean z-scores across H3K4me3, H3K27ac, and H3K36me3, consistent with a background-like regime dominated by weak or absent ChIP-seq signal. At the other extreme, a high-activity composite state shows simultaneously elevated means across multiple marks, capturing bins where promoter-associated and transcription-associated signals co-occur or where broadly active chromatin yields coordinated elevation across tracks. Between these extremes, the remaining states occupy intermediate positions and display mark-selective tendencies, indicating that the model does not merely discretize total signal intensity but separates distinct multivariate combinations of marks.

Because emissions are modeled as conditionally independent Gaussians (diagonal covariance), state identity is expressed primarily through per-mark mean shifts rather than cross-mark covariance structure(7, 8). Practically, this makes Fig. 3 directly interpretable as a set of "emission signatures": each row summarizes how the model expects the three marks to behave in bins assigned to that state. This design choice favors interpretability and parameter stability at the cost of not explicitly modeling residual cross-mark coupling within a state. As a result, differences between states should be read as differences in typical per-mark levels (on the normalized z-score scale), while any remaining dependencies are expected to appear later as residual structure and are therefore handled as diagnostics rather than as part of the core state definition.

The emission patterns also provide biologically meaningful hypotheses prior to external validation.A state characterized by relatively higher H3K4me3 together with elevated H3K27ac but comparatively weaker H3K36me3 is consistent with promoter-proximal activity, where promoter marks are strong but gene-body elongation marks need not dominate. Conversely, a state with relatively stronger H3K36me3—potentially accompanied by moderate H3K27ac—is more compatible with transcription-associated chromatin within gene bodies. The lowest-signal state is a natural background-like candidate. Finally, the composite high-signal state suggests a regime of broadly active chromatin that may include mixtures of active promoters and enhancers, or bins that aggregate multiple regulatory elements within a 200 bp window. In the next section, we show that Viterbi decoding maps these emission programs to long, contiguous genomic segments, and we then use enrichment at CpG islands, TSS, and gene bodies to assign formal functional labels to each state and to resolve any ambiguity between promoter-like and enhancer-like interpretations(7, 9, 10).

**Viterbi decoding produces domain-like segments supported by persistence diagnostics.** We next decoded the most likely state sequence along chr22 using the Viterbi algorithm and visualized representative windows together with the three input tracks(7, 9, 10). Across loci, the decoded labels form extended contiguous segments rather than rapidly switching bin-by-bin (Fig. 4B). This behavior is expected under a first-order HMM when the fitted transition matrix favors state persistence: brief, local fluctuations in one mark are down-weighted unless they are supported by a consistent multivariate shift or sustained evidence across adjacent bins. In practice, the Viterbi path acts as a domain-level summarization of the three-track signal, converting noisy per-bin variation into a small number of stable chromatin "runs" whose boundaries correspond to coordinated changes in the multivariate profile(5–7).

In relatively low-contrast windows, the decoded path can be dominated by a single state over long stretches (Fig. 4A). Importantly, this is not necessarily a failure mode. Such intervals can arise when (i) all three normalized marks remain near their baseline for extended regions, (ii) the window contains limited dynamic range after preprocessing, or (iii) the posterior evidence for alternative states never exceeds the transition penalty required to introduce a new segment. Therefore, interpreting segmentation quality benefits from inspecting both information-rich regions (where multiple marks vary and boundaries are apparent) and low-contrast regions (where a background-like regime should remain stable). This dual inspection guards against over-interpreting flat-looking state tracks as "model collapse" when they may simply reflect genuinely homogeneous chromatin.

To quantify domain-like continuity beyond qualitative inspection, we examined persistence diagnostics derived from the decoded path and the fitted transition dynamics (SI Appendix, Fig. S1)(7, 8). First, the empirical transition matrix is dominated by self-transitions, indicating that once a state is entered

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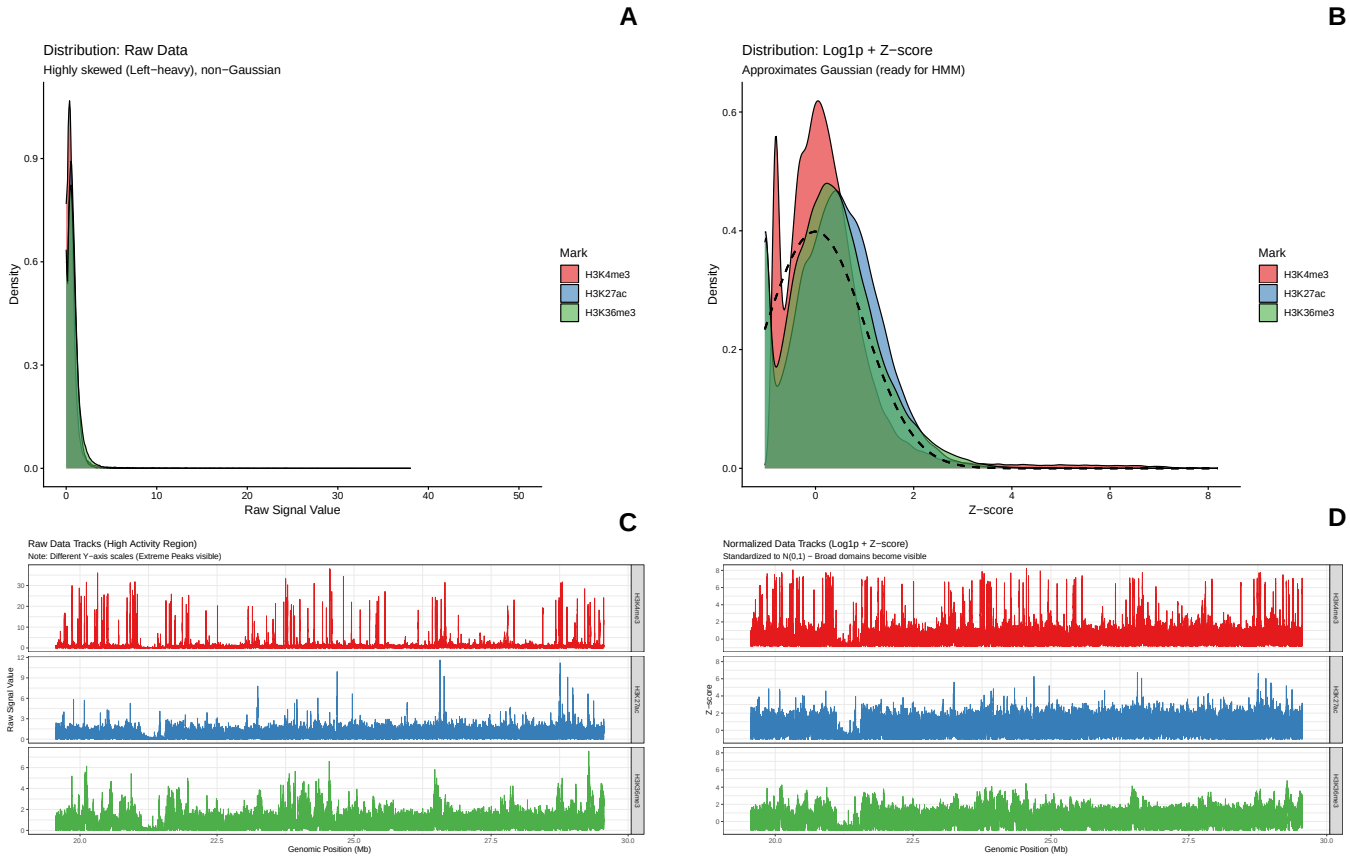


**Fig. 1. Preprocessing reduces extreme-peak dominance and improves Gaussian compatibility of three histone-mark tracks on chr22.** (A) Raw signal density distributions of H3K4me3, H3K27ac, and H3K36me3 are strongly skewed with long tails, indicating that extreme peaks can dominate Gaussian parameter estimation. (B) After log1p transformation and per-mark z-score standardization, distributions become more concentrated and closer to Gaussian-like shapes, improving compatibility with Gaussian-emission HMMs. (C) In a high-activity region, raw tracks have different y-axis scales and extreme peaks visually dominate, obscuring broader patterns. (D) Normalized tracks (log1p + z-score) are on a comparable scale, making broad domains more visible for multivariate modeling.

it tends to persist across neighboring 200 bp bins rather than alternating at high frequency. Second, segment-length distributions span multiple scales, with both short segments and long domains present across states, consistent with chromatin organization operating at heterogeneous genomic scales. Together, these diagnostics support that the HMM is capturing spatial coherence rather than overfitting bin-level noise, and that the four-state solution provides a parsimonious segmentation that is stable over local neighborhoods.

Having established that the inferred states produce coherent genomic domains, we next asked whether these domains correspond to known regulatory landmarks. We therefore evaluated state-wise enrichments at CpG islands, transcription start sites, and gene bodies to convert state indices into functional labels and to resolve promoter-/enhancer-/gene-body-associated interpretations.

**Enrichment at CpG islands, TSS, and gene bodies assigns functional labels to states.** To interpret the four inferred states in functional genomic terms, we computed state-wise fold enrichments over three external annotation sets: CpG islands, transcription start sites (TSS), and gene bodies excluding TSS $\pm$ 2 kb (Fig. 5A–C)(11). These annotations are complementary: CpG islands provide an orthogonal marker of promoter-proximal architecture, TSS directly anchors transcription initiation sites, and gene bodies (with TSS neighborhoods removed) isolate transcription-associated chromatin away from promoter-proximal signal. Using all three avoids over-interpreting any single landmark and enables a more constrained mapping from state indices to regulatory categories.

Across CpG islands, enrichments differ substantially among states (Fig. 5A), indicating that the segmentation captures promoter-proximal versus non-promoter regimes rather than merely reflecting a global "high/low" signal split. Consistently, TSS enrichment is strongly state-dependent (Fig. 5B), allowing identification of a promoter-associated state as the one whose occupancy near TSS is highest relative to its genome-wide prevalence on chr22. Formally, fold enrichment compares the fraction of bins assigned to a state within the annotation to the fraction of that state genome-wide; therefore, enrichment $> 1$ indicates preferential localization at the annotation beyond what would be expected from overall state coverage(5, 6).

After excluding TSS $\pm$ 2 kb to prevent promoter spillover, gene-body enrichment highlights states preferentially occurring within transcribed regions (Fig. 5C). This exclusion is critical because promoter-proximal chromatin can extend into the first exons and proximal introns; without removal, promoter-like states can artifactually inflate "gene-body" enrichment. Under the exclusion, a state showing enrichment $> 1$ in gene bodies provides more specific evidence for transcription-associated chromatin organization, which is expected to co-vary with gene-body marks such as H3K36me3.

Combining emission signatures (Fig. 3) with enrichment patterns (Fig. 5), we assign functional labels to the state

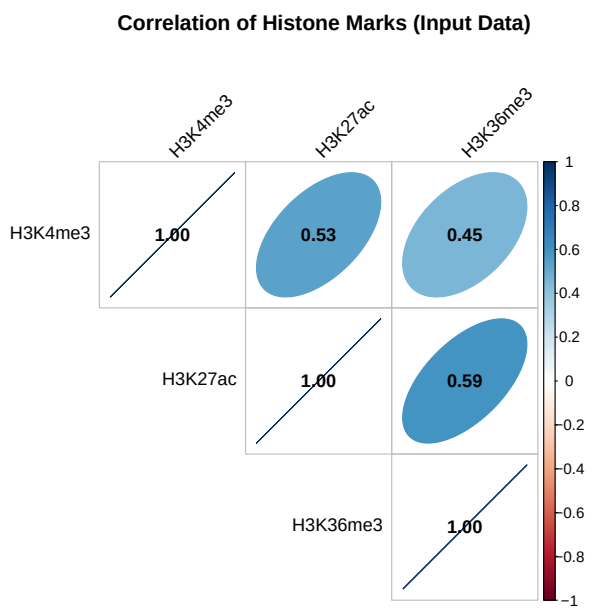


**Fig. 2. Moderate pairwise correlations among input histone marks support joint multivariate state modeling.** Pairwise correlations between normalized H3K4me3, H3K27ac, and H3K36me3 on chr22 are moderate (H3K4me3–H3K27ac = 0.53; H3K4me3–H3K36me3 = 0.45; H3K27ac–H3K36me3 = 0.59), indicating partially overlapping but non-redundant information suitable for multivariate HMM segmentation.

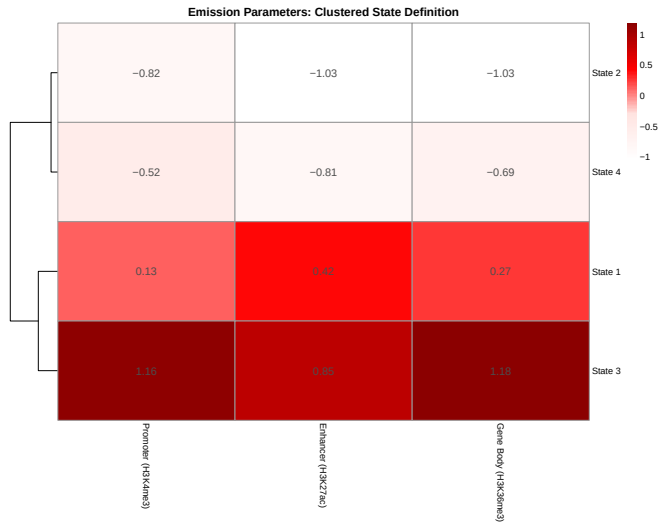


**Fig. 3. Emission means reveal a stratified four-state chromatin organization across three histone marks.** Clustered heatmap of state-wise emission means (z-score scale) for H3K4me3, H3K27ac, and H3K36me3. State 3 shows the highest mean signal across marks (high-activity composite), State 1 is intermediate-high, State 4 is low-to-intermediate, and State 2 is lowest (background-like). Under conditionally independent Gaussian emissions (diagonal covariance), state separation is primarily driven by per-mark mean differences.

indices as follows. The state with the strongest TSS enrichment, supported by elevated CpG-island enrichment, is labeled *promoter-like*. The state preferentially enriched over gene bodies after excluding TSS $\pm$ 2 kb is labeled *gene-body-like*, consistent with transcription-associated chromatin(3). The lowest-signal state, characterized by weak enrichment across all three annotations and low emission means across marks, is labeled *background-like*. The remaining state exhibits high emission means across multiple marks and moderate-to-broad enrichment signals, and we label it *high-activity composite*. This composite label is intentionally conservative: with only three marks, a state can reflect mixtures of active promoters and enhancers (both often show elevated H3K27ac) and cannot always be cleanly separated without additional tracks (e.g., H3K4me1, H3K27me3) or transcript evidence(12).

These label assignments convert the segmentation from numeric indices (State 1–4) into interpretable regulatory categories that can be used to summarize chr22 at the domain level. In the Discussion, we use these functional labels to explain both the biological consistency of the states and practical limitations such as promoter-like assignments in regions without sharp peaks, which can arise from multivariate evidence accumulation and HMM smoothing rather than peak-calling behavior.

Finally, we performed additional diagnostics to evaluate residual structure and to detect degeneracy modes in state-number selection; these analyses are presented in the SI Appendix (Figs. S2–S3) to keep the main narrative focused on the segmentation and its biological interpretation.

## Discussion

The inferred four-state segmentation is biologically interpretable and supported by independent enrichment analyses at CpG islands, TSS, and gene bodies(5, 6). The state emission structure provides a compact summary of multivariate histone-mark patterns, and the HMM transition structure yields contiguous domains that align with the notion of spatially coherent chromatin organization.

A notable behavior is that bins lacking sharp peaks can still be assigned to promoter-like states. This arises naturally from multivariate inference and sequential smoothing: moderate coordinated signal across marks can outweigh any single-track peak, and high self-transition probabilities encourage the model to assign borderline bins to the dominant local segment. In this sense, the HMM output should be interpreted as a domain annotation rather than a peak-calling result(5, 6).

Our preprocessing (log1p and z-scoring) improves numerical stability but has limitations. These transforms do not guarantee Gaussianity and can obscure absolute differences in dynamic range across marks. In datasets with substantial zero inflation, transformed values can cluster at a common mode, contributing to the sharp residual peak near zero. Future work could explore robust scaling (median/MAD), quantile-based transforms, heavier-tailed emission families (e.g., Student-$t$), or mixture/zero-inflated emissions.

In addition, AIC and BIC can track each other closely in this setting because the effective sample size is large at the bin level (thousands of 200-bp bins), making the penalty difference between $2p$ (AIC) and $p \log n$ (BIC) less decisive compared with the magnitude of likelihood changes induced by EM local solutions(13, 14).

A practical question is why both AIC and BIC show an apparent drop once $K$ exceeds $\sim 5$ on the evaluated subset (SI Appendix, Fig. S3). In principle, increasing $K$ strictly expands model flexibility and can improve the maximized

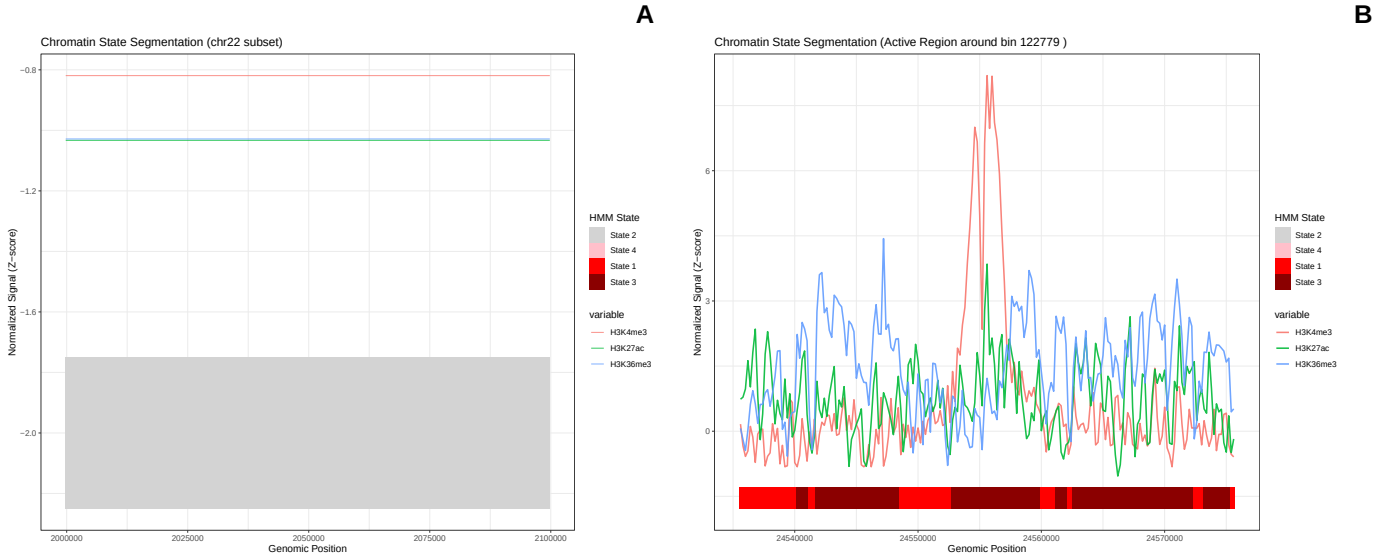


**Fig. 4. Viterbi-decoded HMM states form contiguous chromatin domains aligned with multivariate mark patterns.** (A) Example chr22 subset with relatively low contrast shows near-uniform segmentation dominated by a single state across a long interval, consistent with a genuinely homogeneous (background-like) region and illustrating that uninformative windows can yield visually flat state tracks. (B) In an active region, the decoded state sequence forms extended contiguous segments rather than bin-level switching, demonstrating spatial smoothing: state changes occur at domain boundaries while the three normalized marks vary continuously above the state track.

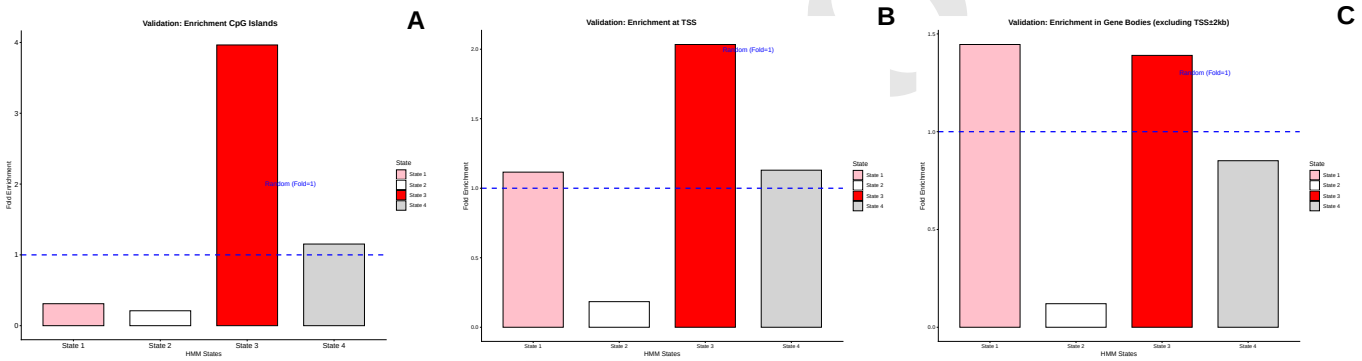


**Fig. 5. Functional enrichment analyses validate promoter- and gene-body-associated state behavior.** (A) State-wise fold enrichment over CpG islands shows differential occupancy across the four inferred HMM states, consistent with promoter-proximal regulatory architecture. (B) Fold enrichment over transcription start sites (TSS) identifies a promoter-like state with elevated enrichment relative to the random baseline (Fold = 1). (C) Fold enrichment over gene bodies excluding TSS $\pm$ 2 kb reveals states preferentially occurring within transcribed regions; enrichment $> 1$ indicates that a state occurs in gene bodies more often than expected from genome-wide coverage alone.

log-likelihood, and both criteria reward such likelihood gains while penalizing added parameters(13, 14). A visible decrease in AIC/BIC at higher $K$ therefore often indicates that the additional states are providing a large likelihood gain relative to the penalty term, but that gain is not necessarily biologically meaningful. In small or low-contrast regions, extra states can act as "variance absorbers" (e.g., carving out a tiny number of extreme bins or capturing slight mean shifts) or enable EM to reach alternative local optima with sharper, more separated Gaussians(15, 16). This can yield a disproportionate improvement in log-likelihood even when the decoded segmentation is unstable or effectively degenerate (e.g., near-single-state occupancy or very few segments), because information criteria do not directly penalize pathological occupancy or lack of spatial diversity.

For this reason, a downward trend beyond $K \approx 5$ should be interpreted together with stability and occupancy diagnostics rather than as decisive evidence that the underlying biology requires more than four states. We treat these diagnostics as safeguards against overconfident state-number selection. In practice, we recommend selecting $K$ based on a combination of interpretability (emission coherence and enrichment-consistent labeling), robustness across random restarts, and rejecting solutions with extreme state occupancies or near-trivial segmentations (SI Appendix, Fig. S3).

## Materials and Methods

**Data and binning.** We analyzed chr22 using three histone marks (H3K4me3, H3K27ac, H3K36me3) represented as continuous bigWig tracks. The chromosome was partitioned into non-overlapping 200 bp bins. For each bin $t$, we computed a 3-dimensional observation vector $\mathbf{x}_t = (x_{t,1}, x_{t,2}, x_{t,3})$ corresponding to the three marks.

**Preprocessing.** For each mark, raw binned signals were transformed by $\tilde{x} = \log(1 + x)$ to reduce the influence of extreme peaks, then standardized to z-scores across bins: $z = (\tilde{x} - \mu)/\sigma$, where $\mu$ and $\sigma$ are the mean and standard deviation of the transformed signal for that mark.

**Hidden Markov model.** We used a $K$-state HMM with latent states $z_t \in \{1, \ldots, K\}$ and first-order Markov transitions $p(z_t \mid z_{t-1}) =$

$A_{z_{t-1},z_t}$. Emissions were modeled as conditionally independent Gaussians across marks:

$$\mathbf{x}_t \mid z_t = k \sim \mathcal{N}(\boldsymbol{\mu}_k,\, \mathrm{diag}(\boldsymbol{\sigma}_k^2)). \qquad [1]$$

Parameters $(\pi, A, \{\boldsymbol{\mu}_k, \boldsymbol{\sigma}_k^2\}_{k=1}^K)$ were estimated by EM (Baum–Welch), iterating (i) forward–backward inference to compute posterior responsibilities and (ii) closed-form updates for transition probabilities and Gaussian parameters. The final segmentation was obtained by Viterbi decoding to compute the most likely state path $\hat{z}_{1:T}$.

**Validation and diagnostics.** We evaluated biological consistency using state-wise fold enrichment over three external annotation sets: CpG islands, transcription start sites (TSS), and gene bodies excluding TSS $\pm$ 2 kb. Let $\mathcal{R} \subseteq \{1, \ldots, T\}$ denote the set of bins overlapping an annotation. Fold enrichment was computed as

$$\mathrm{FE}(k, \mathcal{R}) = \frac{\sum_{t \in \mathcal{R}} \mathbb{I}(\hat{s}_t = k)/|\mathcal{R}|}{\sum_{t=1}^{T} \mathbb{I}(\hat{s}_t = k)/T}. \qquad [2]$$

Diagnostics included (i) mark–mark correlation (Pearson on normalized bins), (ii) empirical transition matrix (from the decoded path), (iii) segment-length distribution (run lengths $\times$ 200 bp), (iv) local genome visualization of tracks plus state calls, and (v) residuals $\mathbf{r}_t = \mathbf{x}_t - \boldsymbol{\mu}_{\hat{s}_t}$.

**Reproducibility.** All analyses were performed in R using Bioconductor utilities for genomic data and `depmixS4` for HMM fitting (code and exact software versions to be provided in the final submission).

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# SI Appendix for: HMM-based chromatin state segmentation on chr22 using three histone marks (H3K4me3/H3K27ac/H3K36me3)

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## SI Appendix

**Overview.** This SI Appendix provides diagnostic analyses that complement the main-text segmentation results. Specifically, we report (i) persistence and segment-length diagnostics that quantify domain-like continuity beyond genome-browser snapshots (Fig. S1), (ii) residual structure under diagonal-Gaussian emissions (Fig. S2), and (iii) a model-selection dashboard illustrating a practical degeneracy mode when evaluating state number on small or low-contrast subsets (Fig. S3). We treat these diagnostics as safeguards against overconfident state-number selection.

### SI Methods: diagnostic definitions and implementation details.

**Empirical transition matrix.** Given the Viterbi-decoded path $\hat{z}_{1:T}$, the empirical transition probability from state $i$ to $j$ is computed as

$$\hat{A}_{ij} = \frac{\sum_{t=1}^{T-1} \mathbb{I}(\hat{z}_t = i,\ \hat{z}_{t+1} = j)}{\sum_{t=1}^{T-1} \mathbb{I}(\hat{z}_t = i)}. \qquad [1]$$

High diagonal values quantify state persistence and support the interpretation of long, coherent chromatin domains.

**Segment-length distribution.** A "segment" is defined as a maximal consecutive run of identical states in $\hat{z}_{1:T}$. Segment length in base pairs is computed as (number of bins) $\times$ 200 bp. Segment-length distributions are reported on a log scale to visualize multi-scale domain structure.

**Residuals under diagonal-Gaussian emissions.** For each bin $t$, residuals are computed as $\mathbf{r}_t = \mathbf{x}_t - \boldsymbol{\mu}_{\hat{z}_t}$ where $\mathbf{x}_t$ is the 3-dimensional normalized observation and $\boldsymbol{\mu}_k$ is the state-wise emission mean vector. A sharp peak near zero indicates that many bins lie close to their assigned state centroid, while heavy tails highlight deviations from Gaussianity, remaining peak-like events, or unmodeled cross-mark dependencies.

**Model-selection dashboard and degeneracy criteria.** We examined information criteria (AIC/BIC) jointly with stability indicators computed on the evaluation subset, including average Viterbi segment length and minimum state coverage (the smallest fraction of bins assigned to any state). A practical degeneracy mode is indicated when the average segment length approaches the evaluated sequence length and the minimum state coverage approaches 100%, consistent with a near-single-state collapse. In practice, this can be mitigated by evaluating on larger/more heterogeneous regions, using multiple random restarts per $K$, and rejecting solutions with extreme state occupancies.

**Fig. S1: persistence and segment-length diagnostics support domain-like continuity.** Figure S1 quantifies the domain-like continuity suggested by genome-browser examples in the main text. The empirical transition matrix is dominated by high self-transition probabilities, indicating that once the model enters a state, it tends to remain there across adjacent bins. Segment-length distributions span multiple orders of magnitude, consistent with a mixture of short and long chromatin domains rather than bin-level label noise.

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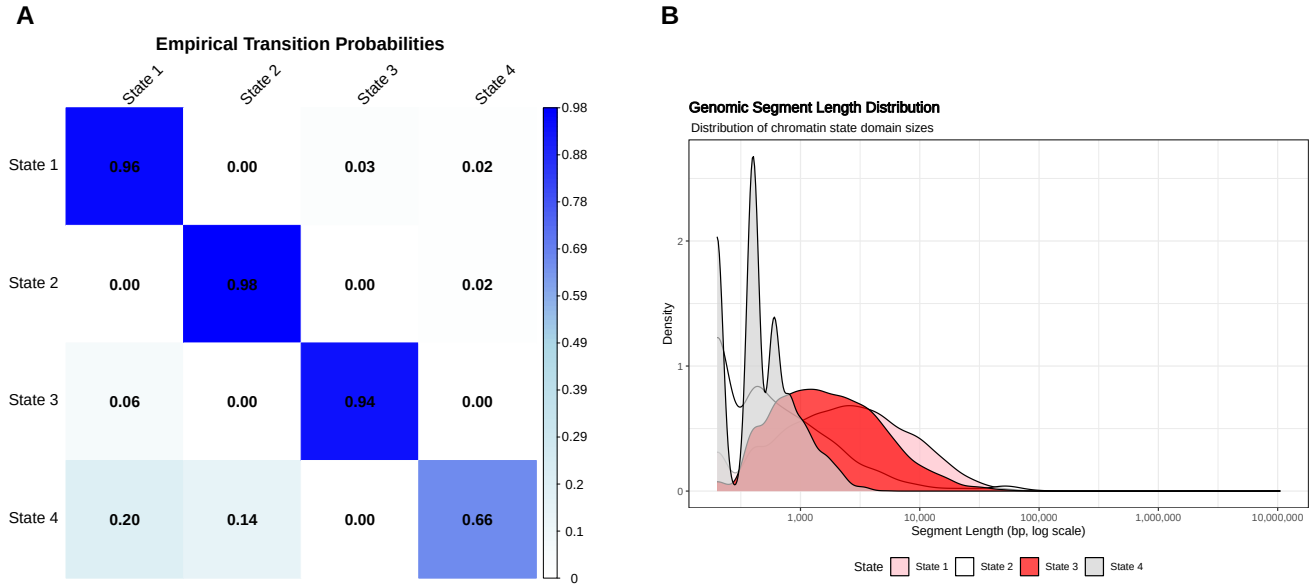


**Fig. 1. Fig. S1. Persistence and segment-length diagnostics of the four-state segmentation.** (A) Empirical transition probabilities estimated from the Viterbi path show strong diagonal dominance, indicating high state persistence across adjacent 200 bp bins. (B) Segment-length distributions (bp, log scale) span multiple orders of magnitude, consistent with multi-scale domain structure rather than rapid bin-level switching.

**Fig. S2: residual distribution highlights well-explained mass with remaining heavy tails.** Figure S2 shows residuals (normalized signals minus state-mean predictions). The prominent mass near zero indicates that many bins are close to the centroid of their assigned state after normalization, while heavy tails reflect deviations from diagonal-Gaussian assumptions (e.g., rare extreme peaks or heterogeneous content within a 200 bp bin). This diagnostic motivates potential extensions such as heavier-tailed emissions (e.g., Student-$t$) or mixture/zero-inflated models when robustness is a priority.

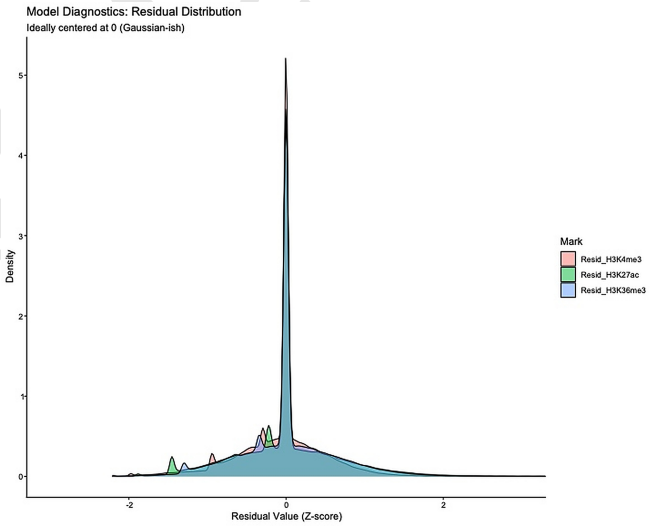


**Fig. 2. Fig. S2. Residual distribution under diagonal-Gaussian emissions.** Residuals are computed as observed normalized signals (z-scores) minus state-mean predictions under the Viterbi-decoded state assignment. A sharp peak near zero indicates many bins lie close to the centroid of their assigned state, while tails reflect deviations from the Gaussian emission assumption and remaining peak-like events.

**Fig. S3: model-selection dashboard illustrates a degeneracy mode on small/low-contrast subsets.** Figure S3 reports AIC/BIC across candidate state numbers and two stability indicators. In this evaluation subset, the average segment length can remain near the full sequence length while minimum state coverage approaches 100%, consistent with a near-single-state collapse. This illustrates why information criteria alone can be misleading on small or low-contrast regions and supports reporting stability-aware diagnostics as safeguards against overconfident state-number selection.

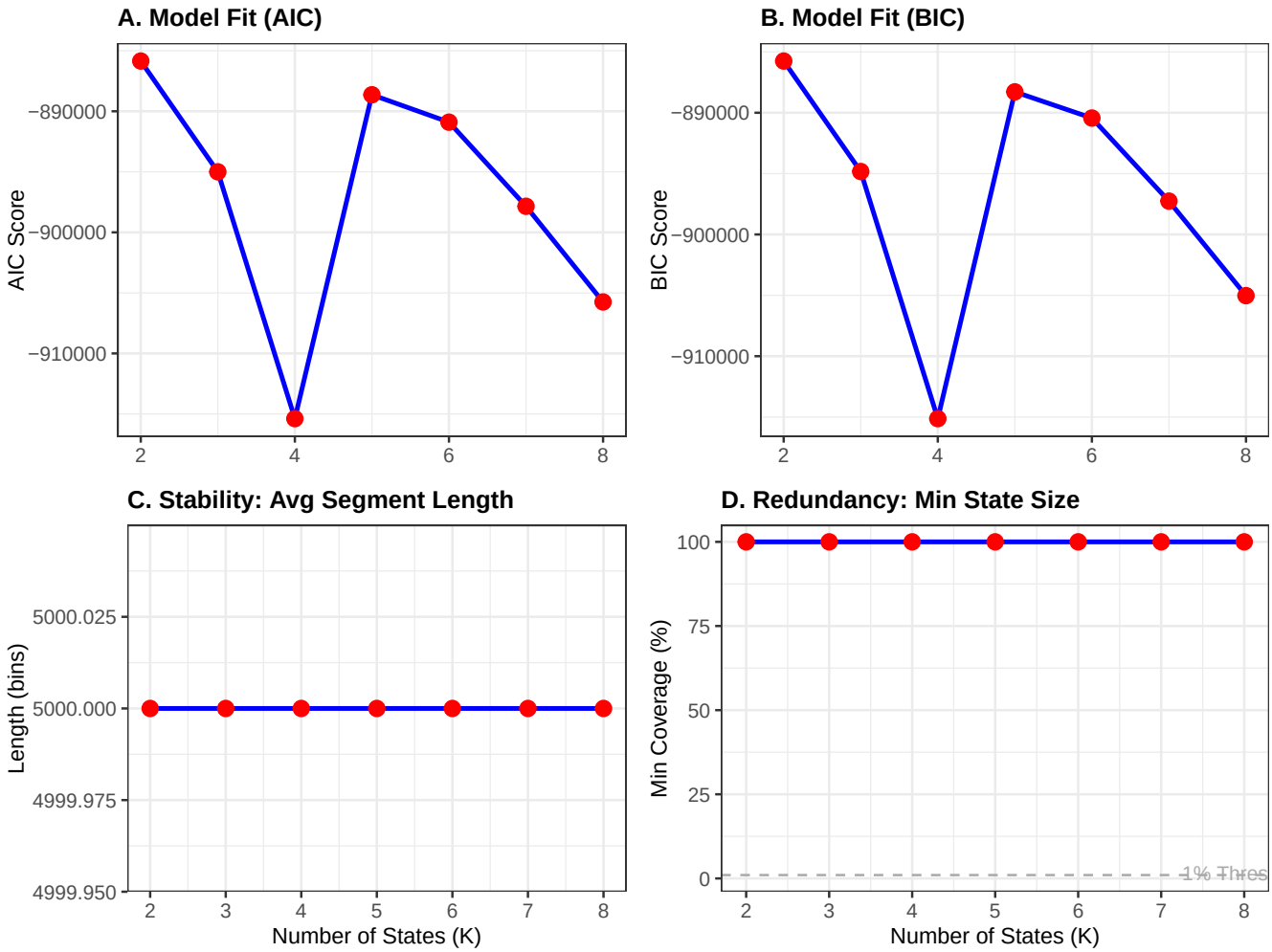


**Fig. 3. Fig. S3. Model-selection dashboard and a practical degeneracy mode.** (A) AIC across candidate state numbers. (B) BIC across the same range. (C) Average Viterbi segment length (bins) on the evaluated subset. (D) Minimum state coverage on the evaluated subset. A pattern of near-constant, sequence-length-scale segments and near-100% minimum coverage is consistent with a near-single-state collapse on that subset, motivating larger-region evaluation and restart-based robustness checks.